GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

Independent Set 5.01 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A 5 cm side becomes 15 cm in the copy. Find the scale factor.

k = 3

2. A 15 cm side becomes 6 cm in the copy. Find the scale factor.

k = 0.4

The copy is smaller, so k is less than 1.

3. A 3 cm by 5 cm rectangle is copied at 7.5 cm by 12.5 cm. Find the scale factor.

k = 2.5

7.5 ÷ 3 and 12.5 ÷ 5 both give 2.5.

4. The scale factor is 1.5 and a side of the original is 8 cm. How long is that side in the copy?

12 cm

5. A map's scale is $1\text{ in} : 25\text{ mi}$. Two cities are 2.8 in apart on the map. How far apart are they?

70 mi

6. On the same map, a road is 150 mi long. How long is it on the map?

6 in

Back to the map: divide by 25.

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

Question Bank 5.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the scale factor from the original to the copy.

1. A 4 cm side becomes 12 cm.

k = 3

2. A 9 cm side becomes 3 cm.

k = ⅓

3. A 2 cm by 7 cm rectangle is copied at 5 cm by 17.5 cm.

k = 2.5

5 ÷ 2 and 17.5 ÷ 7 both give 2.5.

4. A 16 cm side becomes 12 cm.

k = ¾

For questions 5 through 8, use the scale factor or the scale.

5. The scale factor is 4 and a side of the original is 2.5 cm. How long is that side in the copy?

10 cm

6. The scale factor is 0.5 and a side of the original is 9 cm. How long is that side in the copy?

4.5 cm

7. A map's scale is $1\,\text{cm} : 30\,\text{km}$. Two towns are 5.5 cm apart on the map. How far apart are they?

165 km

8. On the same map, a lake is 45 km long. How long is it on the map?

1.5 cm

Back to the map: divide by 30.

Choose one.

9. A drawing has a 65° angle. The real figure is a scale copy at $k = 3$. What is the angle in the real figure?

- ○ A. 68°
- ○ B. 195°
- ● **C. 65°**
- ○ D. 21.7°

C

A scale factor multiplies lengths; angles do not change.

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

Question Bank - Additional 5.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the scale factor from the original to the copy.

1. A 3 cm side becomes 12 cm.

k = 4

2. A 10 cm side becomes 2 cm.

k = $\frac{1}{5}$

3. A 3 cm by 4 cm rectangle is copied at 4.5 cm by 6 cm.

k = 1.5

4.5 ÷ 3 and 6 ÷ 4 both give 1.5.

4. A 20 cm side becomes 15 cm.

k = ¾

For questions 5 through 8, use the scale factor or the scale.

5. The scale factor is 3 and a side of the original is 3.5 cm. How long is that side in the copy?

10.5 cm

6. The scale factor is 0.25 and a side of the original is 12 cm. How long is that side in the copy?

3 cm

7. A map's scale is $1\text{ in} : 40\text{ mi}$. Two cities are 3.25 in apart on the map. How far apart are they?

130 mi

8. On the same map, a river is 90 mi long. How long is it on the map?

2.25 in

Back to the map: divide by 40.

Choose one.

9. A drawing has a 30° angle. The real figure is a scale copy at $k = 4$. What is the angle in the real figure?

- ○ A. 34°
- ○ B. 120°
- ○ C. 7.5°
- ● **D. 30°**

D

A scale factor multiplies lengths; angles do not change.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Independent Set 5.02 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A figure with perimeter 18 cm is scaled by $k = 3$. Find the copy's perimeter.

54 cm

2. A figure with area $7\,\text{cm}^2$ is scaled by $k = 3$. Find the copy's area.

63 cm²

Area × k^2 = 7 × 9.

3. A figure with area $40\,\text{cm}^2$ is scaled by $k = \frac{1}{2}$. Find the copy's area.

10 cm²

Area × (½)² = 40 × ¼.

4. A 2 cm by 5 cm rectangle is scaled by $k = 2.5$. Find the copy's perimeter and area.

P = 35 cm, A = 62.5 cm²

P = 14 × 2.5; A = 10 × 6.25.

5. A figure has area $9\,\text{cm}^2$ and its scale copy has area $144\,\text{cm}^2$. Find k.

k = 4

144 ÷ 9 = 16 = k^2, so k = √16.

6. A figure's perimeter is 12 cm and its copy's perimeter is 48 cm. What is the area ratio, copy to original?

16

k = 4 from the perimeters; the area ratio is k^2 = 16.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 22 cm, $k = 2$.

44 cm

2. Area $5\,\text{cm}^2$, $k = 4$.

80 cm²

5 × 16.

3. Area $72\,\text{cm}^2$, $k = \frac{1}{3}$.

8 cm²

72 × $\frac{1}{9}$.

4. A 3 cm by 7 cm rectangle, $k = 2$. Perimeter and area of the copy.

P = 40 cm, A = 84 cm²

20 × 2; 21 × 4.

For questions 5 through 7, find k , or the ratio asked.

5. Areas $6\,\text{cm}^2$ and $150\,\text{cm}^2$. Find k .

k = 5

150 ÷ 6 = 25 = k^2.

6. Areas $50\,\text{cm}^2$ and $8\,\text{cm}^2$, original to copy. Find k .

k = 0.4

8 ÷ 50 = 0.16 = k^2, so k = 0.4: the copy is smaller.

7. Perimeters 15 cm and 45 cm, original to copy. What is the area ratio?

9

k = 3; k^2 = 9.

Choose one.

8. A figure with area $6\,\text{cm}^2$ is scaled by $k = 5$. What is the copy's area?

- ○ A. $30\,\text{cm}^2$
- ○ B. $60\,\text{cm}^2$
- ● **C.** $150\,\text{cm}^2$
- ○ D. $11\,\text{cm}^2$

C

$6 \times 5^2 = 6 \times 25$.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank - Additional 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 26 cm, $k = 3$.

78 cm

2. Area $9\,\text{cm}^2$, $k = 3$.

81 cm²

9 × 9.

3. Area $100\,\text{cm}^2$, $k = \frac{1}{2}$.

25 cm²

100 × ¼.

4. A 4 cm by 5 cm rectangle, $k = 1.5$. Perimeter and area of the copy.

P = 27 cm, A = 45 cm²

18 × 1.5; 20 × 2.25.

For questions 5 through 7, find k , or the ratio asked.

5. Areas $11\,\text{cm}^2$ and $99\,\text{cm}^2$. Find k .

k = 3

99 ÷ 11 = 9 = k^2.

6. Areas $32\,\text{cm}^2$ and $2\,\text{cm}^2$, original to copy. Find k .

k = ¼

2 ÷ 32 = 1⁄16 = k^2, so k = ¼.

7. Perimeters 10 cm and 50 cm, original to copy. What is the area ratio?

25

k = 5; k^2 = 25.

Choose one.

8. A figure with area $4\,\text{cm}^2$ is scaled by $k = 6$. What is the copy's area?

- ○ A. $48\,\text{cm}^2$
- ● **B.** $144\,\text{cm}^2$
- ○ C. $24\,\text{cm}^2$
- ○ D. $10\,\text{cm}^2$

B

$4 \times 6^2 = 4 \times 36$.

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

Independent Set 5.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 6 in long on it. Find the real length.

12 ft

1 in : 2 ft; 6 × 2.

2. A plan is drawn at $1\text{ in} : 5\text{ ft}$. A room is 3.2 in long on it. Find the real length.

16 ft

3. A plan is drawn at $1\text{ in} : 5\text{ ft}$. A real wall is 22 ft long. Find its length on the plan.

4.4 in

22 ÷ 5.

4. A plan is drawn at $\frac{1}{4}\text{ in} : 1\text{ ft}$. A room is 2 in by 3.5 in on it. Find the real dimensions and area.

8 ft by 14 ft; 112 ft^2

1 in : 4 ft; 2 × 4 and 3.5 × 4; then 8 × 14.

5. A model boat is built at $1 : 40$ and is 15 cm long. Find the real boat's length in meters.

6 m

15 × 40 = 600 cm.

6. On a drawing a wall is 3 in long; the real wall is 16.5 ft. Another wall is 5 in long on the drawing. How long is it really?

27.5 ft

The pair gives 5.5 ft per inch; 5 × 5.5.

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

Question Bank 5.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, use the scale given.

1. $\frac{1}{2}$ in : 1 ft . A wall is 5 in on the plan. Real length?

10 ft

1 in : 2 ft.

2. 1 in : 6 ft . A room is 2.5 in on the plan. Real length?

15 ft

3. 1 in : 6 ft . A real wall is 27 ft. Length on the plan?

4.5 in

27 ÷ 6.

4. $\frac{1}{4}$ in : 1 ft . A room is 1.5 in by 2.5 in on the plan. Real dimensions and area?

6 ft by 10 ft; 60 ft²

1 in : 4 ft; 6 × 10.

For questions 5 through 7, answer in the unit asked.

5. A model train is built at 1 : 80 and is 25 cm long. The real train, in meters?

20 m

25 × 80 = 2,000 cm.

6. On a drawing a wall is 2 in long; the real wall is 15 ft. Another wall is 7 in long on the drawing. How long is it really?

52.5 ft

7.5 ft per inch; 7 × 7.5.

7. A 60-ft boat is modeled at 1 : 60 and at 1 : 90 . Which model is longer, and by how much?

The 1 : 60 model, by 4 in (12 in against 8 in)

60 ft = 720 in; 720 ÷ 60 = 12; 720 ÷ 90 = 8.

Choose one.

8. A plan is drawn at $1\text{ in} : 3\text{ ft}$. A room is 2 in by 4 in on it. What is the room's real area?

- ○ A. $18\,\text{ft}^2$
- ○ B. $36\,\text{ft}^2$
- ○ C. $24\,\text{ft}^2$
- ● **D.** $72\,\text{ft}^2$

D

6 ft by 12 ft; 6 × 12.

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

Question Bank - Additional 5.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, use the scale given.

1. $\frac{1}{2}$ in : 1 ft . A wall is 7 in on the plan. Real length?

14 ft

1 in : 2 ft.

2. 1 in : 8 ft . A room is 1.5 in on the plan. Real length?

12 ft

3. 1 in : 8 ft . A real wall is 36 ft. Length on the plan?

4.5 in

36 ÷ 8.

4. $\frac{1}{4}$ in : 1 ft . A room is 2.5 in by 3.5 in on the plan. Real dimensions and area?

10 ft by 14 ft; 140 ft^2

1 in : 4 ft; 10 × 14.

For questions 5 through 7, answer in the unit asked.

5. A model plane is built at 1 : 60 and is 30 cm long. The real plane, in meters?

18 m

30 × 60 = 1,800 cm.

6. On a drawing a wall is 4 in long; the real wall is 22 ft. Another wall is 7 in long on the drawing. How long is it really?

38.5 ft

5.5 ft per inch; 7 × 5.5.

7. A 36-ft bus is modeled at 1 : 36 and at 1 : 54 . Which model is longer, and by how much?

The 1 : 36 model, by 4 in (12 in against 8 in)

36 ft = 432 in; 432 ÷ 36 = 12; 432 ÷ 54 = 8.

Choose one.

8. A plan is drawn at $1\text{ in} : 5\text{ ft}$. A room is 2 in by 3 in on it. What is the room's real area?

- ○ A. $25\,\text{ft}^2$
- ○ B. $30\,\text{ft}^2$
- ● **C.** $150\,\text{ft}^2$
- ○ D. $50\,\text{ft}^2$

C

10 ft by 15 ft; 10 × 15.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

Converting Length Across Systems; Currency

Independent Set 5.06 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 12 in?

30.48 cm

2. $1\text{ cm} = 0.39\text{ in}$. How many inches is 40 cm?

15.6 in

40 × 0.39 (or 40 ÷ 2.54 = 15.7).

3. $1\text{ mi} = 1.61\text{ km}$. How many kilometers is 10 mi?

16.1 km

4. $1\text{ km} = 0.62\text{ mi}$. How many miles is 8 km?

4.96 mi

8 × 0.62.

5. $1\text{ yd} = 3\text{ ft}$, $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 2 yd? Estimate first.

About 180; exactly 182.88 cm

2 yd = 72 in; 72 × 2.54.

6. Today's rate: $1\text{ US dollar} = 0.92\text{ euros}$. How many euros is $50? How many dollars is 23 euros?

46 euros; $25

50 × 0.92; 23 ÷ 0.92.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

Converting Length Across Systems; Currency

Question Bank 5.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\text{ in} = 2.54\text{ cm}$. 8 in to centimeters.

20.32 cm

2. $1\text{ cm} = 0.39\text{ in}$. 60 cm to inches.

23.4 in

60 × 0.39 (or 60 ÷ 2.54 = 23.6).

3. $1\text{ mi} = 1.61\text{ km}$. 20 mi to kilometers.

32.2 km

4. $1\text{ km} = 0.62\text{ mi}$. 15 km to miles.

9.3 mi

15 × 0.62.

For questions 5 through 7, show every step.

5. $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. 5 ft to centimeters.

152.4 cm

60 in × 2.54.

6. Today's rate: $1\text{ US dollar} = 1.35\text{ Canadian dollars}$. $40 to Canadian dollars; 27 Canadian dollars to US dollars.

54 CAD; $20

40 × 1.35; 27 ÷ 1.35.

7. Rates today: $1\text{ US dollar} = 1.32\text{ Canadian dollars}$; $1\text{ Canadian dollar} = 16.2\text{ pesos}$. $50 to pesos, by way of Canadian dollars.

1,069.20 pesos

50 × 1.32 = 66 CAD; 66 × 16.2.

Choose one.

8. $1\,\text{km} = 0.62\,\text{mi}$. A road is 20 km long. How many miles is that?

- ● **A. 12.4 mi**
- ○ B. 32.2 mi
- ○ C. 1.24 mi
- ○ D. 20.62 mi

A

20 × 0.62.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

Converting Length Across Systems; Currency

Question Bank - Additional 5.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\text{ in} = 2.54\text{ cm}$. 9 in to centimeters.

22.86 cm

2. $1\text{ cm} = 0.39\text{ in}$. 80 cm to inches.

31.2 in

80 × 0.39 (or 80 ÷ 2.54 = 31.5).

3. $1\text{ mi} = 1.61\text{ km}$. 30 mi to kilometers.

48.3 km

4. $1\text{ km} = 0.62\text{ mi}$. 25 km to miles.

15.5 mi

25 × 0.62.

For questions 5 through 7, show every step.

5. $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. 4 ft to centimeters.

121.92 cm

48 in × 2.54.

6. Today's rate: $1\text{ US dollar} = 0.92\text{ euros}$. $30 to euros; 69 euros to US dollars.

27.60 euros; $75

30 × 0.92; 69 ÷ 0.92.

7. Rates today: $1\text{ US dollar} = 1.32\text{ Canadian dollars}$; $1\text{ Canadian dollar} = 16.2\text{ pesos}$. $25 to pesos, by way of Canadian dollars.

534.60 pesos

25 × 1.32 = 33 CAD; 33 × 16.2.

Choose one.

8. $1\,\text{mi} = 1.61\,\text{km}$. A road is 40 mi long. How many kilometers is that?

- ○ A. 24.8 km
- ○ B. 6.44 km
- ● **C. 64.4 km**
- ○ D. 41.61 km

C

40 × 1.61.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 8–9

Converting Mass, Area and Volume

Independent Set 5.08 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $1\,\text{lb} = 0.454\,\text{kg}$. How many kilograms is 30 lb?

13.62 kg

(30 ÷ 2.204 = 13.61 by the other ratio.)

2. $1\,\text{kg} = 2.204\,\text{lb}$. How many pounds is 25 kg?

55.1 lb

3. $1\,\text{gal} = 3.785\,\text{L}$. How many liters is 6 gal?

22.71 L

4. $1\,\text{ft} = 12\,\text{in}$. How many square inches is $9\,\text{ft}^2$?

1,296 in²

9 × 144: the ratio squared.

5. $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. How many liters is 24 fl oz?

0.72 L

3 c × 0.24.

6. $1\,\text{ft} = 0.305\,\text{m}$. How many square meters is $30\,\text{ft}^2$? (Round to two places.)

About 2.79 m²

30 × 0.305²: the sheet's length ratio, squared, across systems.

Converting Mass, Area and Volume

Question Bank 5.08 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\,\text{lb} = 0.454\,\text{kg}$. 20 lb to kilograms.

9.08 kg

(20 ÷ 2.204 = 9.07 by the other ratio.)

2. $1\,\text{kg} = 2.204\,\text{lb}$. 15 kg to pounds.

33.06 lb

3. $1\,\text{gal} = 3.785\,\text{L}$. 8 gal to liters.

30.28 L

4. $1\,\text{L} = 0.264\,\text{gal}$. 25 L to gallons.

6.6 gal

For questions 5 through 7, say how many times the ratio is used.

5. $1\,\text{yd} = 3\,\text{ft}$. $4\,\text{yd}^2$ to square feet.

36 ft²

4 × 9: twice.

6. $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. 32 fl oz to liters.

0.96 L

4 c × 0.24.

7. $1\,\text{ft} = 12\,\text{in}$. $3\,\text{ft}^3$ to cubic inches.

5,184 in³

3 × 1,728: three times.

Choose one.

8. $1\,\text{ft} = 12\,\text{in}$. A rug is $6\,\text{ft}^2$ in area. How many square inches is that?

- ● **A.** $864\,\text{in}^2$
- ○ B. $72\,\text{in}^2$
- ○ C. $144\,\text{in}^2$
- ○ D. $150\,\text{in}^2$

A

6 × 144.

Converting Mass, Area and Volume

Question Bank - Additional 5.08 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\text{ lb} = 0.454\text{ kg}$. 40 lb to kilograms.

18.16 kg

(40 ÷ 2.204 = 18.15 by the other ratio.)

2. $1\text{ kg} = 2.204\text{ lb}$. 5 kg to pounds.

11.02 lb

3. $1\text{ gal} = 3.785\text{ L}$. 12 gal to liters.

45.42 L

4. $1\text{ L} = 0.264\text{ gal}$. 50 L to gallons.

13.2 gal

For questions 5 through 7, say how many times the ratio is used.

5. $1\text{ yd} = 3\text{ ft}$. 7 yd^2 to square feet.

63 ft²

7 × 9: twice.

6. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. 40 fl oz to liters.

1.2 L

5 c × 0.24.

7. $1\text{ ft} = 12\text{ in}$. 4 ft^3 to cubic inches.

6,912 in³

4 × 1,728: three times.

Choose one.

8. $1\,\text{m} = 100\,\text{cm}$. A floor is $3\,\text{m}^2$ in area. How many square centimeters is that?

- ○ A. $600\,\text{cm}^2$
- ○ B. $10{,}000\,\text{cm}^2$
- ○ C. $300\,\text{cm}^2$
- ● **D.** $30{,}000\,\text{cm}^2$

D

3 × 10,000.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

Independent Set 5.10 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A car gets 32 miles per gallon and its tank holds 14 gal. How far does it go on a tank?

448 mi

2. That car makes a 1,000-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

3 tanks

1,000 ÷ 448 ≈ 2.23 → 3.

3. A recipe for 4 servings uses $1\frac{1}{4}$ cups of rice. How much rice for 10 servings?

3⅛ cups

k = 2.5; 2.5 × 1¼.

4. $1\,\text{km} = 0.62\,\text{mi}$. A train does 120 km/h; a bus does 70 mi/h. Which is faster? Show the speeds in one unit.

The train: 120 km/h = 74.4 mi/h > 70 mi/h

120 × 0.62.

5. Juice is mixed orange : mango : water = $3:2:5$. A batch uses 9 cups of orange. How many cups is the whole batch?

30 cups

k = 3; 10 parts × 3.

6. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$; a room is 2 in by 3.5 in on it. Find the room's real area.

252 ft²

12 ft by 21 ft. (The square-meter version is the test's transfer item and is not practised here — ruling 18.)

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

Question Bank 5.10 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, one rate, used as many times as it takes.

1. A car gets 25 miles per gallon with a 16-gal tank. Miles on a full tank?

400 mi

2. That car makes a 1,500-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

4 tanks

1,500 ÷ 400 = 3.75 → 4.

3. A printer prints 42 pages per minute. How long does a 1,260-page job take?

30 min

For questions 4 through 6, the same unit first.

4. $1\,\text{mi} = 1.61\,\text{km}$. Runner A: 3 mi in 24 min. Runner B: 5 km in 25 min. Each speed in km/h, and who is faster?

A ≈ 12.08 km/h, B = 12 km/h; A

4.83 ÷ 0.4; 5 ÷ (5⁄12).

5. A recipe for 6 servings uses $2\frac{1}{4}$ cups of flour. Flour for 16 servings? With 9 cups on hand, how many servings?

6 cups; 24 servings

k = 8⁄3; 9 ÷ 2¼ = 4 recipes × 6.

6. Paint is mixed $1:3:4$. A batch uses 6 gal of the second color. The whole batch?

16 gal

k = 2; 8 parts × 2.

Choose one.

7. A car gets 30 miles per gallon with a 10-gal tank. It makes a 750-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

- ● **A. 3**
- ○ B. 25
- ○ C. 75
- ○ D. 2

A

300 mi per tank; 750 ÷ 300 = 2.5 → 3.

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

Question Bank - Additional 5.10 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, one rate, used as many times as it takes.

1. A car gets 24 miles per gallon with a 15-gal tank. Miles on a full tank?

360 mi

2. That car makes a 1,300-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

4 tanks

1,300 ÷ 360 ≈ 3.6 → 4.

3. A printer prints 36 pages per minute. How long does a 900-page job take?

25 min

For questions 4 through 6, the same unit first.

4. $1\,\text{km} = 0.62\,\text{mi}$. Car A: 90 km/h. Car B: 60 mi/h. Which is faster? Show both in mi/h.

B: 90 km/h = 55.8 mi/h < 60

90 × 0.62.

5. A recipe for 8 servings uses 3 cups of flour. Flour for 12 servings? With 9 cups on hand, how many servings?

4½ cups; 24 servings

k = 1.5; 9 ÷ 3 = 3 recipes × 8.

6. Paint is mixed $2:3:5$. A batch uses 15 gal of the third color. The whole batch?

30 gal

k = 3; 10 parts × 3.

Choose one.

7. A car gets 35 miles per gallon with a 12-gal tank. It makes a 1,050-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

- ○ A. 30
- ● **B. 3**
- ○ C. 2
- ○ D. 87.5

B

420 mi per tank; 1,050 ÷ 420 = 2.5 → 3.

GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · MA.7.GR.1.5 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

k = 1.6

copy ÷ original

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

k = 0.25

the copy is smaller, so k is under 1

3. A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

9 cm

4. A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ● **D. 65°**

D

5. A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

135 km

drawing × rate

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · MA.7.GR.1.5 · 6 questions

6. A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

40 cm

perimeter × k

7. A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

56.25 cm²

area × k^2 = 9 × 6.25

8. A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

a. Find the copy's perimeter.

10 cm

20 × ½

b. Find the copy's area.

6 cm²

24 × ¼

9. A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find k.

k = 4

160 ÷ 10 = 16 = k^2

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

9

k = 3 from the perimeters; the area ratio is k^2

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.

the reason and the numbers

PART C. Real-World Scale Drawings

Lesson 4 · MA.7.GR.1.5 · 4 questions

12. A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

15 ft

1 in : 2 ft; 7.5 × 2

13. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

3.5 in

real ÷ rate

14. A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

8 m

32 × 25 = 800 cm

15. SSDD. Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a length conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

a. Find the real length of the wall drawn 3 in long.

12 ft

1 in : 4 ft; 3 × 4

b. Find the room's real area.

192 ft²

12 by 16; the rate used twice

c. Write the scale factor from the drawing to the room as a single number.

48

1 in : 4 ft is 1 in : 48 in

d. Find the room's real perimeter in meters.

17.08 m

2(12 + 16) = 56 ft; 56 × 0.305 (56 ÷ 3.28 = 17.07 by the other ratio). The area in square meters is the test's transfer item and is not rehearsed here (ruling 18).

PART D. Converting Length Across Systems; Currency *Lessons 6–7 · MA.7.AR.3.3 · 6 questions*

16. $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 14 in?

35.56 cm

17. $1\text{ cm} = 0.39\text{ in}$. How many inches is 120 cm?

46.8 in (47.2 by ÷ 2.54)

fewer inches than centimeters

18. $1\text{ km} = 0.62\text{ mi}$. How many miles is 30 km?

18.6 mi

19. $1\text{ yd} = 3\text{ ft}$, $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 6 yd? Estimate first.

About 540; exactly 548.64 cm

216 in × 2.54

20. Rate on Nov 3: $1\text{ US dollar} = 0.92\text{ euros}$.

a. How many euros is $150?

138 euros

× the rate

b. How many US dollars is 69 euros?

$75

÷ the rate

21. $1\text{ in} = 2.54\text{ cm}$; $1\text{ cm} = 0.39\text{ in}$. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

Lee used the ratio whose 1 is inches on centimeters. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).

the error named and the estimate

PART E. Converting Mass, Area and Volume

Lessons 8–9 · MA.7.AR.3.3 · 5 questions

22. $1\text{ lb} = 0.454\text{ kg}$. How many kilograms is 80 lb?

36.32 kg

(80 ÷ 2.204 = 36.30 by the other ratio)

23. $1\text{ L} = 0.264\text{ gal}$. How many gallons is 40 L?

10.56 gal

24. $1\text{ ft} = 0.305\text{ m}$. How many square meters is 8 ft^2 ? (Round to two places.)

About 0.74 m^2

$8 \times 0.305^2 = 0.7442$: the sheet's length ratio, squared

25. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. How many liters is 48 fl oz?

1.44 L

6 c × 0.24

26. $1\text{ m} = 3.28\text{ ft}$. A desk is 1.5 m^2 in area. Which is closest to its area in square feet?

- ○ A. 4.9 ft^2
- ○ B. 9.8 ft^2
- ● **C.** 16.1 ft^2
- ○ D. 10.8 ft^2

C

$1.5 \times 3.28^2 = 16.1376$.

PART F. Multi-Step Proportion Problems

Lessons 10–11 · MA.7.AR.4.5 · 4 questions

27. A car gets 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

348 mi

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks (800 ÷ 348 ≈ 2.3)

a part of a tank is still a tank

28. A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

a. How much rice for 12 servings?

5 cups

k = 4

b. With 10 cups on hand, how many servings can be made?

24 servings

8 recipes × 3

29. $1\,\text{km} = 0.62\,\text{mi}$. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

The train: 110 km/h = 68.2 mi/h > 65 mi/h

same unit first

30. Paint is mixed $1:2:5$. A batch uses 10 gal of the third color. How many gallons is the whole batch?

16 gal

k = 2; 8 parts × 2

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

A, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k^2 — 36 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². What is the scale factor? *1 pt*

k = 3

72 ÷ 8 = 9 is k^2, so k = √9. A board with 9 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

12 ft

the scale per 1 inch, then ×: 1 in : 4 ft

b. Find the room's real area. *1 pt*

168 ft²

real dimensions first: 12 by 14 — 10.5 in^2 × 4 = 42 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 9.30 m² (3.05 m by 3.05 m); 9.29 by the sheet's other ratio

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28^2 = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

22.86 cm (23.1 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

A and C

Both sheet routes (27.3 and 27.6) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb (44.05 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

18.925 L (18.94 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ● **C. 1.86 m^2**
- ○ D. 12.2 m^2

C

20 × 0.305^2 = 1.8605 (or 20 ÷ 3.28^2 = 1.86).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k = 2.5 from the yields, then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

4 recipes × the yield; 24 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

B, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 40 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

k = 4

80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then ×: 1 in : 2 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

27.94 cm (28.2 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

B and D

Both sheet routes (35.1 and 35.4) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

457.2 cm

a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg (27.22 by ÷ 2.204)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

34.065 L (34.09 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 m = 3.28 ft. A floor is 5 m^2 in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 ft^2**
- ○ B. 16.4 ft^2
- ○ C. 10.8 ft^2
- ○ D. 32.8 ft^2

A

5 × 3.28^2 = 53.79 (or 5 ÷ 0.305^2 = 53.75).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 72.90 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k = 1.5 from the yields, then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

4 recipes × the yield; 48 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

32 gal

k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

k = 3

copy ÷ original

Scoring: 1 point.

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☒ **B. Each side is 2 times as long.**
- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

B, C and D

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

k = 5

150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

10 ft

the scale per 1 inch, then ×: 1 in : 5 ft

b. Find the room's real area. *1 pt*

150 ft²

real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

135 ft²

9 ft by 15 ft.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

17.78 cm (17.9 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

B and D

Both sheet routes (42.9 and 43.3) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

137.16 cm

a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

77.14 lb (77.09 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

11.355 L (11.36 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 in = 2.54 cm. A tile is 10 in^2 in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 cm^2
- ● **B. 64.5 cm^2**
- ○ C. 6.45 cm^2
- ○ D. 50.8 cm^2

B

$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

118.80 Canadian dollars

× the rate

b. How many US dollars is 66 Canadian dollars? *1 pt*

$50

÷ the rate; 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

270 mi

the rate × the gallons

b. It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

5 tanks (1,200 ÷ 270 ≈ 4.44)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

6 cups

k = 3 from the yields, then the ingredient

b. With 10 cups on hand, how many servings can be made? *1 pt*

25 servings

5 recipes × the yield; 50 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

24 gal

k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.

Same unit first; 80 > 45 compared two units.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.